

electric potential

Links closely with [gravitational potential](#).

Electric potential is a property of **an electric field**, not of individual objects within the field. It is a quantity that is **independent** of the magnitude of the charge on the object on which the force acts.

Electric potential is a **scalar** quantity.

Electric potential V_e in a radial field is given by $V_e = \frac{kq}{r}$ where q is the charge that **produces** the radial field. This follows from [Coulomb's law](#) and the relationship between field strength and potential.

Electric potential energy is the energy (work) required to bring two point charges q_1 and q_2 **from infinity** to a separation of r ; $E_p = k \frac{q_1 \times q_2}{r}$.

There is zero potential at infinity.

The work done W in moving a charge q in an electric field is $W = q \times \Delta V_e$, where ΔV_e is the **electric potential difference**.

The unit of electric potential is $\text{J C}^{-1} \equiv V$.

When there are many separate charges, each one has to be brought from ∞ , so the total electric potential energy is the **sum of each separate energy transfer**.

$$\text{electric field strength } E = -\frac{\Delta V_e}{\Delta r}$$

[charged conducting spheres](#) provide a weird ass example of electric field strength and electric potential doing things.